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To: Prof. Gentry
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Subject: EMS 180 - HW5

Problem 1

Cost equation:

$$Cost = C_{\text{materials}} + C_{\text{dedicated}} + C_{\text{capital}} + C_{\text{overhead}}$$

Where:

$$C_{\text{materials}} = C_m * \frac{m}{1-f}$$

$$C_{\text{dedicated}} = \frac{C_t}{n} \text{ceiling}(\frac{n}{n_t})$$

$$C_{\text{capital}} = \frac{L_f}{\dot{n}} \frac{C_c}{t_{wo}}$$

$$C_{\text{overhead}} = \frac{\dot{C}_{oh}}{\dot{n}}$$

- 1.1 List each of the numbers that will be used as inputs to the cost equation. You should provide a brief explanation of how you selected your variables from the ranges provided in the software.**

Capital cost (C_c)	180,000 dollars	Actual cost of equipment[1]
Material utilization factor ($1 - f$)	0.935	Average from CES EduPack
Production rate ($\dot{n}$)	0.5 hours	Average from [2]
Tooling cost (C_t)	1500 dollars	Average from [3]
Tool life (n_t)	5000	Average from CES EduPack
Material cost (C_m)	2.27 dollars/kg	Average from CES EduPack

List of references:

1. Teyun's 1100Tons Al Extrusion Press Machine
2. HOONLY website <https://haluminium.com/Product/analysis-on-production-process-of-6063-alloy>
3. Profile Precision Extrusion website <https://profileprecisionextrusions.com/common-questions-answers>

- 1.2 Calculate the cost per frame if 5,000 frames are manufactured. You must show your work or provide information on where this comes from.**

additional parameters:

$$m = 0.150\text{kg}$$

$$C_{oh} = 100 \text{ dollars/hr}$$

$$t_{wo} = 5 \text{ years}$$

$$n = 5000\text{parts}$$

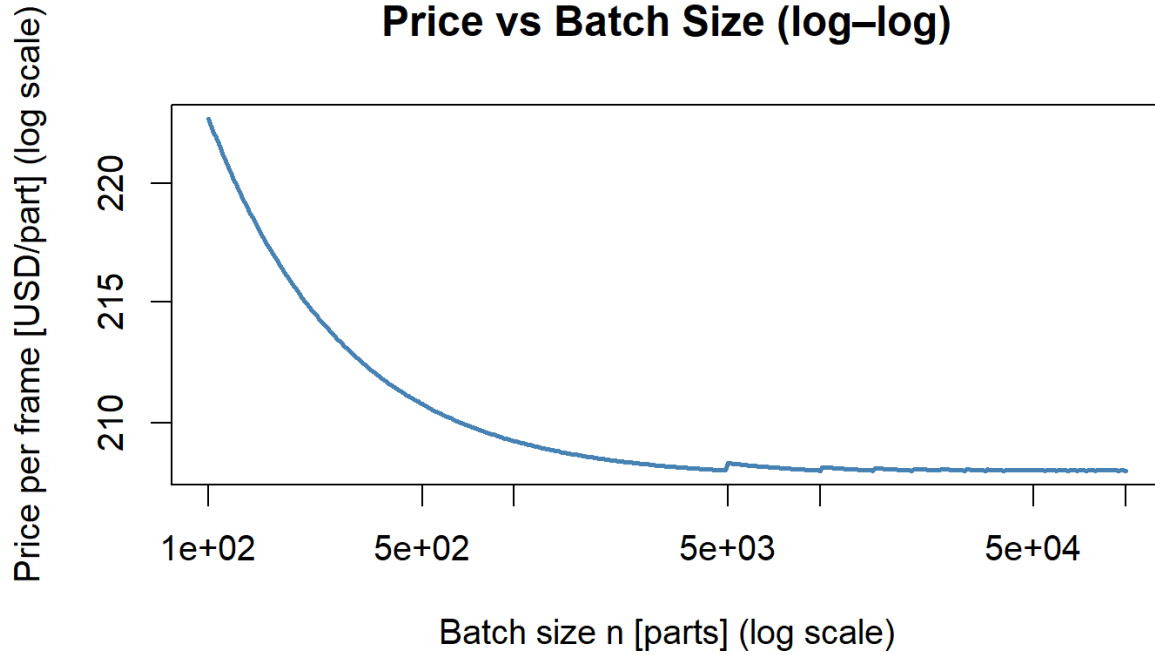
$$L_f = 0.9$$

$$Cost = C_m \frac{m}{1-f} + \frac{C_t}{n} \text{ceiling}(\frac{n}{n_t}) + \frac{L_f}{\dot{n}} \frac{C_c}{t_{wo}} + \frac{\dot{C}_{oh}}{\dot{n}}$$

$$Cost = 2.27 \frac{0.15}{0.935} + \frac{1500}{5000} \text{ceiling}\left(\frac{5000}{5000}\right) + \frac{0.9}{0.5} \frac{180000}{5\left(\frac{12*4*7*24}{1}\right)} + \frac{100}{0.5}$$

$$Cost = 208.7 \text{ dollars}$$

- 1.3 On log-log axes, plot the price versus batch size for the frames, spanning batch sizes of at least 100-100,000 frames. You must use a plotting program (such as Excel or MATLAB), NOT CES EduPack.



$$Cost = \left(C_m \frac{m}{1-f} + \frac{L_f}{\dot{n}} \frac{C_c}{t_{wo}} + \frac{\dot{C}_{oh}}{\dot{n}}\right) + \frac{C_t}{n} \text{ceiling}\left(\frac{n}{n_t}\right)$$

$$\log(Cost) = \text{const.} + \log\left(\frac{C_t}{n}\right) + \log \text{ceiling}\left(\frac{n}{n_t}\right)$$

R language is used to create the plot.

Generative AI is used for plot aesthetics.

1.4 Code snippet used:

```
Cm  <- 2.27
m   <- 0.150
f   <- 0.065
Ct  <- 1500
nt  <- 5000
Lf  <- 0.9
Cc  <- 180000
two_hr <- 5 * 365 * 24
Coh <- 100
n_dot <- 0.5

cost_per_part <- function(n) {
  Cm * m / (1 - f) +
    (Ct / n) * ceiling(n / nt) +
    (Lf / n_dot) * (Cc / two_hr) +
    (Coh / n_dot)
}
## GEN AI CODE START
# Batch sizes and costs
n_vals  <- round(exp(seq(log(100), log(1e5), length.out = 400)))
price_usd <- cost_per_part(n_vals)

# Log-log plot
plot(n_vals, price_usd,
     log = "xy", type = "l", lwd = 2, col = "steelblue",
     xlab = "Batch size n [parts] (log scale)",
     ylab = "Price per frame [USD/part] (log scale)",
     main = "Price vs Batch Size (log{log})")
```